

Detecting Citation Drift in AI-Generated Literature Summaries

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Abstract

Citation drift in AI-generated literature summaries refers to the divergence between a generated claim and the source it cites, including fabricated references, corrupted bibliographic metadata, overgeneralized interpretation, and misattribution of evidential support. This problem has become salient as large language models (LLMs) are increasingly used for literature review, related-work drafting, and evidence synthesis, while remaining prone to hallucination and factual inconsistency [1–3]. Empirical studies show that AI systems can generate plausible-looking but nonexistent citations at substantial rates, with variation by model, domain, and prompt framing; even when cited works are real, metadata errors and unsupported attribution remain common [4–8]. This paper synthesizes research on citation distortion, factual inconsistency in summarization, retrieval-augmented generation, and automated citation verification to define citation drift, organize its error modes, and review emerging detection methods. The evidence indicates that reliable detection requires multi-stage pipelines combining bibliographic resolution, evidence retrieval, claim decomposition, and entailment-style verification, while current benchmarks and metrics remain fragmented and domain-limited [9–12]. The most important shift is from checking whether a citation exists to checking whether it faithfully supports a local claim. The biggest open question is how to build scalable, domain-robust, fine-grained evaluation systems for long-form scholarly synthesis.

Keywords: Citation drift; citation hallucination; large language models; literature summarization; retrieval-augmented generation; citation integrity; factual consistency; scholarly NLP.

1 Introduction

Scientific summarization aims to condense a growing research literature while preserving meaning, salience, and evidential support [13–15]. In practice, LLMs are now used for literature search, abstract screening, related-work drafting, and long-form review generation because manual synthesis is increasingly burdensome as publication volume expands [16–19]. Yet these systems remain vulnerable to hallucination, factual inconsistency, and opaque attribution, which makes them risky for scholarly synthesis [2, 16, 20].

“Citation drift” is a useful umbrella term for the ways a generated summary can separate from its evidence base. It includes at least three distinct failures: fabricated or non-resolving references, bibliographic mutation of real references, and semantic drift in which the cited source exists but does not support the generated claim as stated [4, 12, 21, 22]. This framing connects recent AI

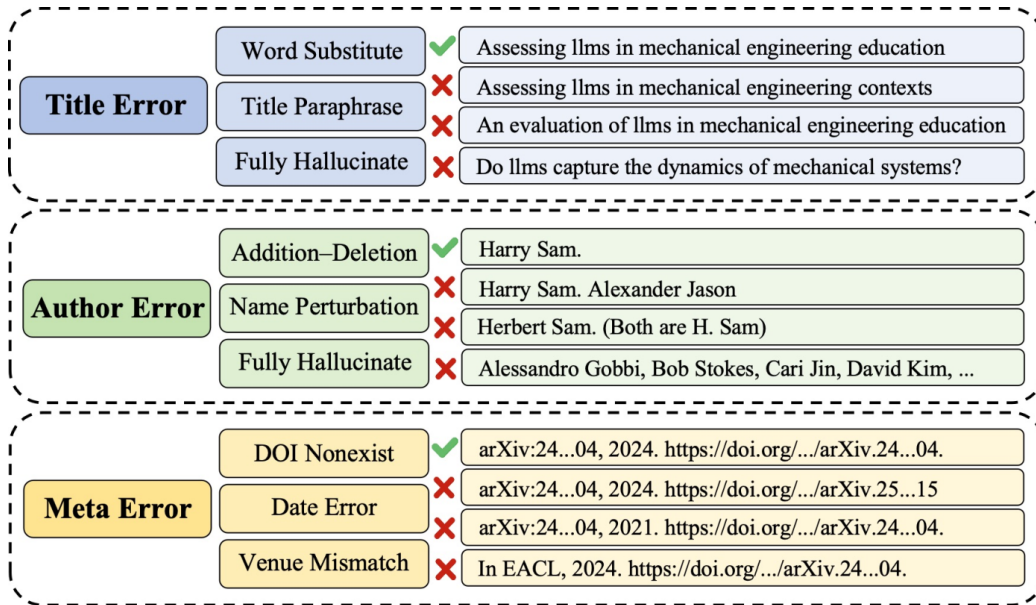


Figure 1: Taxonomy of citation hallucination types.

citation-hallucination work to older meta-research on quotation error, citation distortion, and the propagation of unsupported claims through scientific networks [23–25].

2 Background and Related Work

Human scholarship already exhibits substantial citation distortion, providing a conceptual baseline for understanding AI-generated drift. In biomedical and psychological literatures, studies of quotation and miscitation report nontrivial rates of unsupported, overgeneralized, or contradictory citing claims. One review estimated a 14.5% quotation error rate in medical research, while a large psychology study found that about 19% of citing claims omitted nuance or mischaracterized findings [24, 25]. Citation bias also systematically favors positive findings, which can amplify unsupported beliefs over time [26].

Claim-specific citation-network studies show how bias, amplification, and invention can transform tentative or refuted claims into apparent facts through repeated citation alone [23]. Similar dynamics appear in modern case studies, including persistent post-retraction citation and pandemic-era misrepresentation of null ophthalmology findings. These examples show that citation drift is not merely a language-generation defect but also reflects a broader scholarly integrity problem [30, 31].

3 Main Technical Discussion

Detecting citation drift is harder than ordinary factuality checking because it is inherently multi-document and relational. Citation integrity requires linking a citing claim to the correct reference article, identifying the relevant evidence span within that article, and judging whether the article

Table 1: Types of citation drift in AI-generated literature summaries.

Drift Type	Operational Description	Representative Evidence
Fabrication	Citation points to no real scholarly work	High fabricated-reference rates in ChatGPT and Bard outputs [4, 5]
Metadata mutation	Real target paper exists, but title, authors, year, DOI, or venue are corrupted	DOI hallucinations and partial matches are common [12, 21]
Attribution drift	Real citation does not support the generated claim	Generated citations often lack abstract-level substantiation [27]
Interpretive drift	Summary broadens or alters study scope or conclusion	LLM summaries overgeneralize scientific conclusions [28]
Network drift	Repetition diffuses unsupported claims through later writing	Citation cascades and post-retraction propagation sustain misinformation [23, 29]

actually supports, contradicts, or fails to address the claim [9]. Citation-based scientific summarization and claim verification are therefore adjacent but incomplete antecedents: the former links citations to source text without assessing accuracy, and the latter often ignores the citing-document context and bibliographic layer [9].

Current evidence suggests that bibliographic validity and semantic support must be separated analytically. Bibliographic checks ask whether a reference exists and whether its fields match authoritative records; semantic checks ask whether the cited source warrants the generated proposition [10–12]. This distinction matters because major hallucinations are often easier to detect than minor mutations or support mismatches [12].

4 Current Approaches and Developments

Empirical audits show large variation in citation reliability across models, disciplines, and prompts. GPT-3.5 fabricated 55% of citations in one multidisciplinary study, versus 18% for GPT-4, while substantive errors remained common even among real references [4]. In systematic-review prompting, hallucination rates ranged from 28.6% for GPT-4 to 91.4% for Bard, with low precision and recall against gold-standard review references [5]. Domain studies report similarly unstable performance in psychiatry, ophthalmology, librarianship, finance, mental health, and student writing [6–8, 37–39].

Recent systems increasingly use retrieval augmentation to reduce drift. RAG is motivated by the need to ground generation in external sources and improve verifiability [42–44]. Benchmarks such as ALCE evaluate correctness and citation quality jointly, showing that even strong systems often lack complete citation support roughly half the time on difficult tasks [1]. More recent grounded-generation methods report stronger attribution accuracy, including sentence-level approaches near 90% citation accuracy and self-reflective retrieval methods that improve citation accuracy in long-form generation [35, 45, 46].

Table 2: Pipeline for detecting citation drift.

Stage	Goal	Common Methods	Main Limitation
Citation resolution	Determine whether the citation maps to a real work	Database lookup, fuzzy matching, field-level comparison [11]	Brittle to noisy or incomplete strings [11]
Evidence retrieval	Find candidate passages supporting the claim	Scholarly retrieval, reranking, multi-source search [9, 11]	Retrieval misses relevant support or surfaces
Claim attribution	Align local claim spans with evidence spans	Atomic claim decomposition, sentence-level citation generation [34, 35]	distractors [32, 33] Coarse paragraph-level citations mask local drift [34, 35]
Support judgment	Decide whether evidence supports the claim	NLI, fact verification, structured LLM judging [9, 10, 36]	Evaluators still miss subtle inconsistencies [9, 33]

Table 3: Recent systems for citation drift detection.

System	Core Idea	Reported Strength
Citation Integrity corpus/model	Retrieve evidence sentences, then classify citation accuracy	Best model reached 0.59 micro-F1; subtle errors remain hard [9]
CiteGuard	Retrieval-aware citation attribution alignment	68.1% on CiteME, near human 69.2% [40]
CiteAudit	Multi-agent bibliographic verification benchmark	Stronger verification than commercial and open baselines [11]
CiteCheck	Retrieve candidate work, then classify exact/minor/major drift	88.7 macro-F1 and 88.9% accuracy on physics benchmark [12]
String-feature classifier	Detect hallucinated references without external retrieval	AUC 0.876 cross-validation; useful as prescreening [41]

Automated survey-generation systems are also beginning to bake in citation controls. LITERAS, OpenScholar, ScholarCopilot, AutoSurvey2, and DeepSurvey all combine retrieval with iterative or agentic verification and report improved citation fidelity relative to unguided generation [27, 47–50]. But retrieval alone does not solve attribution drift, because models still struggle with information integration, negative rejection, and false-information handling in RAG settings [33].

5 Challenges, Research Gaps, and Future Directions

The first challenge is conceptual: “citation accuracy” is often underdefined. Existing studies variously measure citation existence, metadata validity, DOI correctness, relevance to a topic, or support for a claim, which makes results hard to compare across systems [5, 21, 51]. Benchmarks for scientific citation generation and verification remain fragmented across sentence-level, document-level, and long-form tasks [34, 52].

Table 4: Research gaps in citation drift detection.

Priority Gap	Why It Matters	Plausible Direction
Fine-grained claim-level benchmarks	Current metrics miss local attribution drift	Build datasets with atomic claims, source spans, and support labels [33, 34]
Domain-robust verification	Performance drops on long-tail topics	Evaluate across humanities, biomedicine, and specialized subfields [8, 21]
End-to-end scholarly pipelines	Retrieval alone does not ensure faithful synthesis	Combine retrieval, decomposition, support judgment, and revision loops [32, 50]
Publisher-side safeguards	Drift can enter preprints and publications	Automated screening, full-text deposit, and disclosure requirements [4, 8, 53]

The second challenge is evaluation. Standard summarization metrics do not capture factual consistency or citation support well, and automatic metrics often correlate weakly with expert judgments in high-stakes summarization [54–56]. LLM-based evaluators are promising, but current factuality judges remain imperfect, especially in domain-specific settings such as clinical summarization [33, 57].

The third challenge is generalization across domains and prompt conditions. Citation fabrication rates vary by discipline, topic familiarity, and specificity, with poorer performance in underrepresented, long-tail, or specialized domains [8, 21, 27]. Lower-visibility or newer topics appear especially vulnerable to hallucination and metadata corruption [7, 16].

The fourth challenge is workflow integration. Manual validation remains necessary in evidence synthesis and AI-assisted review generation, and real-world adoption of automation is still limited outside screening tasks [18, 58]. Because citations can function as data in reviews and bibliometric studies, hallucinated references can cross the line from poor scholarship into possible research misconduct when left unchecked [22].

6 Conclusion

Detecting citation drift in AI-generated literature summaries is not a single verification task but a layered problem spanning reference existence, metadata integrity, evidence retrieval, and claim-level support. The field’s most important shift is from auditing whether a citation looks real to auditing whether each generated claim is faithfully grounded in an identifiable source. The biggest open question is how to scale that fine-grained verification across domains, long contexts, and multi-paper syntheses without making automated scholarly assistance too slow, brittle, or costly.

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